

SALES FORECASTING

SARIMA TIME-SERIES FORECASTING

Full Analysis, Model Output, Forecast & Recommendations

1. Executive Summary

This project develops a monthly sales forecasting solution using the SARIMA (Seasonal AutoRegressive Integrated Moving Average) framework. The notebook loads a 9,800-row sales dataset, converts order dates into a monthly sales series, checks stationarity, separates the last 12 months for testing, fits a SARIMA(1,1,1)x(1,1,1,12) model, evaluates the forecast, and then refits the model on the complete monthly series to generate a 12-month forecast for 2019.

The validation results recorded in the notebook are MAE = 13,930.02, RMSE = 16,394.82, and MAPE = 27.77%. The final model produces 12 monthly forecasts for 2019, with forecast values ranging from 40,285.47 to 113,342.88.

2. Dataset Overview

Attribute	Value
Rows	9,800
Columns	18
Historical period used for monthly series	January 2015 – December 2018
Monthly observations	48
Training observations	36
Testing observations	12
Target variable	Sales
Country represented	United States

The dataset includes order, customer, geography, product, category and sales fields. For forecasting, the analysis focuses on Order Date and Sales.

3. Data Preparation & Monthly Aggregation

The notebook converts Order Date into a time-series index and resamples Sales at monthly frequency using the sum of sales within each month. This creates a 48-month monthly sales series from 2015 through 2018.

The first five monthly observations recorded in the notebook are: January 2015 = 14,205.707; February 2015 = 4,519.892; March 2015 = 55,205.797; April 2015 = 27,906.855; May 2015 = 23,644.303.

4. Exploratory & Time-Series Analysis

The notebook plots historical monthly sales, a 12-month rolling mean, and an additive seasonal decomposition with a period of 12. It also identifies potential high-demand periods using a threshold defined as mean monthly sales plus one standard deviation.

The use of a 12-month seasonal period is consistent with the monthly nature of the series and allows the SARIMA model to capture annual seasonal behavior.

5. Stationarity Analysis

The Augmented Dickey-Fuller (ADF) test was applied to the original monthly series. The recorded ADF statistic was -4.4161 with p-value 0.000278. The notebook also applies first differencing and records an ADF statistic of -8.7271 with p-value 3.27e-14 for the differenced series.

6. SARIMA Model Development

The forecasting model is implemented with statsmodels' SARIMAX class. The selected configuration is SARIMA(1,1,1)x(1,1,1,12), where the seasonal period is 12 months.

Component	Parameter	Value
Non-seasonal	p	1
Non-seasonal	d	1
Non-seasonal	q	1
Seasonal	P	1
Seasonal	D	1
Seasonal	Q	1
Seasonal period	s	12

The training sample contains 36 observations and the test sample contains 12 observations. The fitted model reports AIC = 202.966, BIC = 203.952 and HQIC = 200.838.

Important model-output caveat: the notebook reports that the covariance matrix is singular or near-singular, with a very large condition number, and warns that standard errors may be unstable. This should be considered when interpreting individual coefficient significance.

7. Model Diagnostics & Output

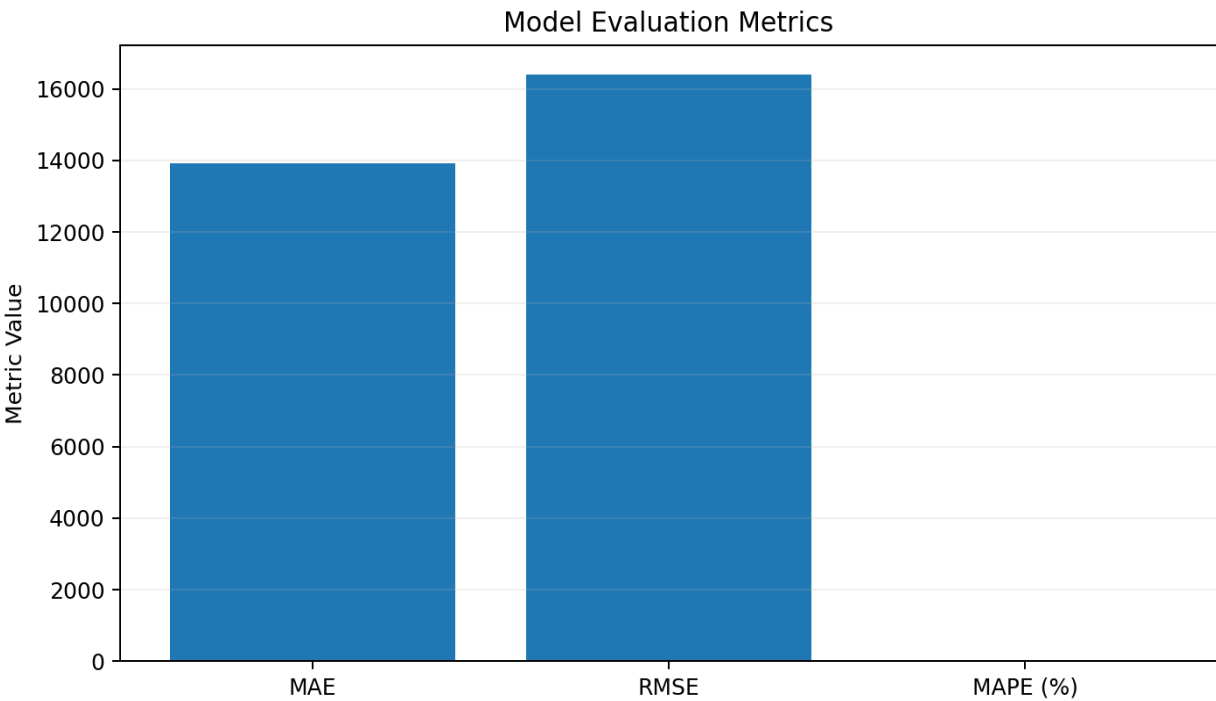
Diagnostic	Recorded output
Log Likelihood	-96.483
AIC	202.966
BIC	203.952
HQIC	200.838
Ljung-Box Q (L1)	0.00

Prob(Q)	0.99
Jarque-Bera	0.41
Prob(JB)	0.82
Heteroskedasticity H	1.26
Prob(H)	0.85

The recorded Ljung-Box result has Prob(Q)=0.99, while the Jarque-Bera result has Prob(JB)=0.82. These are diagnostic outputs from the notebook and should be interpreted together with the model warning about the covariance matrix.

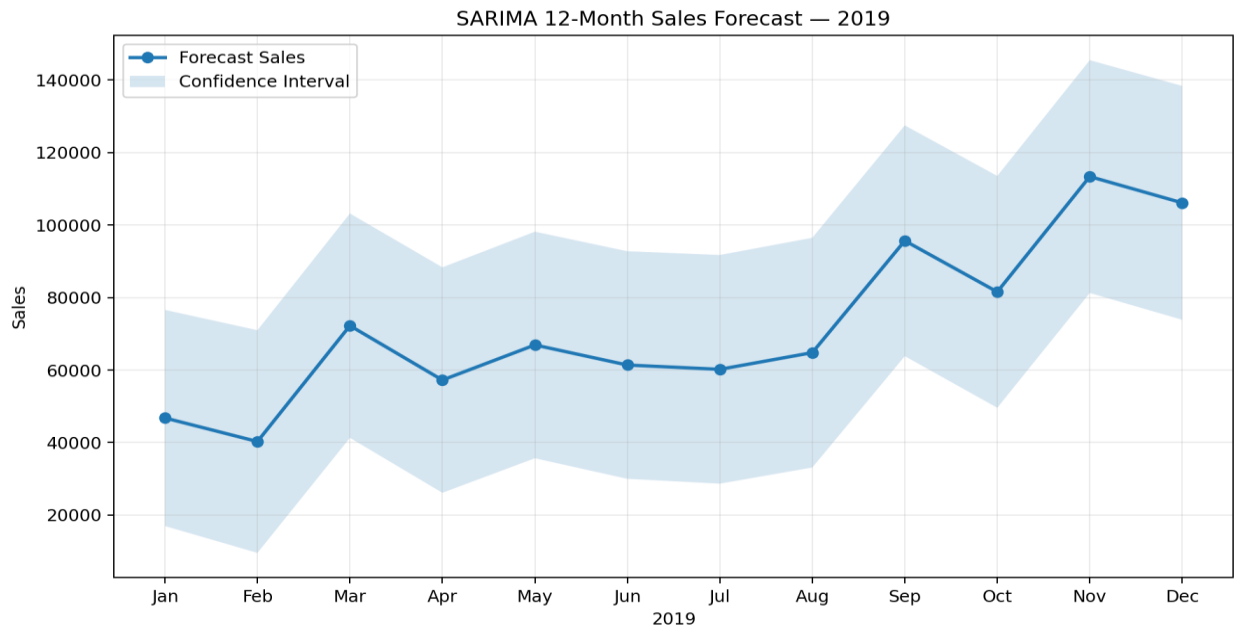
8. Forecast Evaluation

Metric	Value	Interpretation
MAE	13,930.02	Average absolute forecast error.
RMSE	16,394.82	Penalizes larger forecast errors more strongly.
MAPE	27.77%	Average absolute percentage error.



9. 2019 Future Sales Forecast

The final SARIMA model is refitted on the complete monthly sales series and used to forecast 12 months ahead. The total of the twelve point forecasts is approximately 866,448.34, with an average monthly forecast of 72,204.03. The highest forecast is 113,342.88 in Nov and the lowest is 40,285.47 in Feb.



Month	Forecast Sales	Lower Bound	Upper Bound
Jan	46,782.48	16,992.67	76,572.28
Feb	40,285.47	9,596.04	70,974.90
Mar	72,234.20	41,318.86	103,149.55
Apr	57,236.03	26,161.99	88,310.07
May	66,910.83	35,688.42	98,133.24
Jun	61,378.25	30,009.64	92,746.87
Jul	60,193.67	28,679.73	91,707.62
Aug	64,798.24	33,139.62	96,456.86
Sep	95,640.89	63,838.19	127,443.58
Oct	81,551.12	49,604.93	113,497.31

Nov	113,342.88	81,253.76	145,432.00
Dec	106,094.28	73,862.78	138,325.79

10. Business Insights

- The model produces a clear monthly forecast profile rather than a single annual estimate, supporting month-by-month planning.
- Forecast demand is weakest in Feb (40,285.47) and strongest in Nov (113,342.88).
- The forecast increases materially from Sep onward and reaches the highest level in November, indicating that year-end planning is particularly important.
- The confidence intervals are wide for several months, so operational decisions should use forecast ranges rather than treating the point forecast as a guaranteed sales value.
- The validation MAPE of 27.77% indicates that forecast error remains material; the model is useful for directional and planning support, but should not be treated as a high-precision demand estimate without further improvement.

11. Recommendations

- Use the monthly forecast as a planning baseline for inventory, procurement and sales targets.
- Increase preparedness for the stronger forecast months, particularly September through December, while avoiding excessive inventory in lower-forecast periods.
- Use the confidence intervals to define conservative and aggressive planning scenarios rather than relying only on the point forecast.
- Monitor actual monthly sales against forecast and retrain the model periodically as new observations become available.
- Improve model selection through systematic SARIMA parameter search and time-series cross-validation.
- Investigate external drivers such as promotions, discounts, holidays, product mix and regional demand if those variables are available.
- Because the notebook reports a near-singular covariance matrix warning, validate model stability and consider alternative specifications before using coefficient-level inference for high-stakes decisions.

12. Conclusion

The project successfully demonstrates an end-to-end monthly sales forecasting workflow using SARIMA. The analysis transforms transaction-level sales into a monthly time series, evaluates stationarity, uses a seasonal SARIMA configuration, validates the forecast against a 12-month holdout period, and generates a 12-month future forecast for 2019. The resulting forecast provides a practical planning baseline, while the reported error metrics and model-stability warning highlight the need for ongoing validation and refinement.

13. Technical Stack

Python • Pandas • NumPy • Matplotlib • Seaborn • Statsmodels • Scikit-learn • Google Colab